Extra Activity - 2

The due date for submitting this assignment has passed.
Due on 2026-03-30, 23:59 IST.
You may submit any number of times before the due date. The final submission will be considered for grading.

You have last submitted on: 2026-03-20, 08:51 IST

In this activity, we will take a look at the application of geogebra to visualize linear transformations $T: R^2 \rightarrow R^2$ T : R2 → R2 . The notation in the questions that follow will be as in the following project

https://www.geogebra.org/m/krqstvdm

The description of the elements in the project is as follows:

• The parameters a, b, c a, b, c and d d are variables used to determine the linear transformation whose matrix representation with respect to the standard bases is $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$[acbd]

• $Te1$ T e1 and $Te2$ T e2 are vectors which represent the images of the standard basis vectors $e1$ e1 and $e2$ e2, respectively.

• u is an arbitrary vector in the plane, and Tu T u is its image under T T .

• P P is a quadrilateral in the plane and TP TP is its image under T T .

• S S is a circular sector in the plane and TS TS represents its image under T T .

To specify the transformation, one can specify the values of a, b, c a, b, c and d d. Or alternatively, move the vectors $Te1$ T e1 and $Te2$ T e2 to specify the image of the standard basis, thus determining the linear transformation. Once a linear transformation is fixed, the vector u can be moved by dragging its arrowhead. The vector Tu T u changes accordingly to give the image of the vector u u. Similarly, the elements P P and S S can be moved around and TP TP and TS TS can be analyzed to understand the geometric properties of the linear transformation. Answer the following questions using the given project.

1 point

Consider the linear transformation T which scales each of the standard basis vectors by a non-zero constant. In the geogebra project, this corresponds to moving the vectors T e1 and T e2 so that the arrowheads are on $(\lambda, 0)$(λ, 0) and $(0, \lambda)$(0, λ), respectively, for some $\lambda \in R$ λ ∈ R \ {0} of your choice.

Which of the following statements are true for such a linear transformation?

- [x] The matrix of T with respect to the standard bases for the domain and codomain is a scalar matrix.
- [x] If λ < 0, the angle between u and T(u) is 180° 180◦ , for all u $\in R^2$ R2 \ {(0, 0)}.
- [ ] If P is the square with vertices (0, 0),(1, 0),(1, 1) and (0, 1), then its image T P has area equal to λ square units.
- [x] The linear transformation T is always a bijection.
- [ ] If λ > 0, then there is a non-zero vector u such that T(u) = u.

Yes, the answer is correct.

Score: 1

Accepted Answers:

The matrix of T with respect to the standard bases for the domain and codomain is a scalar mat
If λ < 0, the angle between u and T(u) is 180° 180◦ , for all u $\in R^2$ R2 \ {(0, 0)}.
The linear transformation T is always a bijection.

It is known that the matrix of a linear transformation T which rotates every vector counter-clockwise by an angle of θθ is given by

$$\begin{bmatrix} cos\theta & -sin\theta \\ sin\theta & cos\theta \end{bmatrix}$$[cosθsinθ−sinθcosθ]

In the geogebra project, set

a = 0.6, b = −0.8, c = 0.8, d = 0.6

This gives a transformation which rotates every vector in the plane. Consider the sector SS whose vertices are (0, 0),(1, 2) and (2, 1). Under the above transformation, the image of SS is a sector TSTS, which is rotated counter-clockwise by an angle sin^{-1} (0.8)sin−1(0.8). If the vertices of TSTS are (0, 0),(−1, 2) and (0.4, k), find the value of kk.

2.2

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 2.2

1 point

Move the head of the vector T e1 to (1, −1). Observe the image T P of the polygon P as you vary the vector T e2 such that its head lies on the line x = −2. There is exactly one point (−2, y_0 y0) such that T P is a line segment. For all other points, the image of P is a quadrilateral. Find the value of y_0 y0.

2

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 2

1 point

1 point

Consider the transformation in the previous question for which the image of P is a line segment. Which of the following statements are true about the transformation?

- [x] The image of the transformation is a line through the origin.
- [ ] The set {(1, 1)} is a basis for the Kernel of the transformation.
- [ ] The matrix of the transformation is invertible.
- [x] There is exactly one line through the origin whose image under the transformation is the singleton set {(0, 0)}.
- [x] The set {(−1, 1)} is a basis for the image of the transformation.

Yes, the answer is correct.

Score: 1

Accepted Answers:

The image of the transformation is a line through the origin.
There is exactly one line through the origin whose image under the transformation is the singleton set {(0, 0)}.
The set {(−1, 1)} is a basis for the image of the transformation.